

Room 4 (P4 D-algebraicity): moderator summary

2026-09-26

Two seats (Tao, Erdős), both UNREVIEWED at the seat level but with extra care taken given that this exact question already produced one retracted claim (P4_v2_note.tex).

Tao's redo of the pole-rate summation: BOUND STILL ONLY LINEAR-TYPE, this time correctly derived. Using the actual general formula from P1f_lemma.tex ($\Delta V = (\pi i s/N)(L_n + i\pi s/N)$, u_j spaced $2\pi/|\Delta V|$ apart) together with the uniform bound $|L_\mu| \leq 1.727$ for all $N \geq 16$ (lem:LS(b)), the correct scaling is $|\Delta V| = \Theta(1/N)$, hence $1 - |q_j^*(\zeta)| = \Theta(1/(jN))$: the per- ζ coefficient *shrinks* like $1/N$, not the constant-in- N or growing-in- N behaviour either the original retracted attempt or this room's working hypothesis considered. Summing over $N \leq M$ gives an aggregate coefficient of only $\Theta(M)$ – still linear growth in the cutoff, not superpolynomial, and two further gaps (non-uniform $j_0(\zeta)$, non-uniform “ r large enough” threshold) would likely weaken this further. This *confirms*, with an actual derivation, what the earlier retraction's correction paragraph asserted but did not derive – no new false claim, no progress either.

Erdős's alternative-obstruction search: NOTHING WORKED, precisely diagnosed. Three elementary candidates (functional-equation degree incompatibility, Wronskian/algebraic-independence search, Cauchy–Kovalevskaya parameter counting) were checked. Each either needs exactly the missing quantitative growth/counting ingredient P4_note.tex already names, or is structurally blocked by results paper2a already proves in the opposite direction (prop:noqdiff, cor:Qminusone rule out the functional relation a degree-counting argument would need).

Room verdict: NO PROGRESS, honestly. Both seats independently converge on the same conclusion: the missing ingredient (a quantitative, superpolynomial-strength pole-counting or growth bound) is not reachable by the routes tried, and no alternative obstruction was found either. This is not a failure of the room – it correctly avoided repeating the

earlier error, and it closes off two more candidate routes with precise reasons rather than leaving them dangling. P4 stands exactly where `P4_note.tex` (v10.19.0) left it: OPEN, with the right theorem family named and the same two missing ingredients.